

## CSCE 753 COMPUTER VISION · MODULE 04 · LECTURE NOTES

# Two-View Geometry

*What two pictures determine.*

## MODULE OUTCOMES

- Derive the epipolar constraint.
- Distinguish the fundamental and essential matrices.
- Recover relative pose and triangulate.
- Explain the homography and when it applies instead.
- Diagnose the degenerate configurations.

## 1 The epipolar constraint

### TWO CAMERAS AND A WORLD POINT ARE COPLANAR, AND THAT IS THE WHOLE CONSTRAINT

**The two camera centres and the world point define a plane** — the epipolar plane — and both image rays, being rays from those centres to that point, lie within it.

**So three vectors are coplanar: the ray in the first camera, the ray in the second, and the baseline between the centres.** Coplanarity of three vectors is precisely the vanishing of their triple product, so the constraint is  $\mathbf{x}' \cdot (\mathbf{t} \times \mathbf{R} \mathbf{x}) = 0$ .

**Rewriting the triple product as a matrix gives  $\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0$  with  $\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R}$** , where  $[\mathbf{t}]_{\times}$  is the skew matrix representing the cross product. **The essential matrix is the triple product rearranged**, and seeing it that way removes most of its mystery.

**The consequence is the useful part. A point in one image constrains its match to a line in the other** — the epipolar line, the projection of the first ray into the second image. **Correspondence search drops from two dimensions to one**, which is exactly what makes dense stereo computationally feasible (Module 06) and what makes sparse matching far more reliable once an initial pose is known.

|             | Fundamental matrix $\mathbf{F}$                         | Essential matrix $\mathbf{E}$                                                           |
|-------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Operates on | Raw pixel coordinates.                                  | Calibrated (normalised) image coordinates — pixels premultiplied by $\mathbf{K}^{-1}$ . |
| Requires    | Nothing. No calibration at all.                         | Known intrinsics (Module 02).                                                           |
| Relation    | $\mathbf{F} = \mathbf{K}'^T \mathbf{E} \mathbf{K}^{-1}$ | $\mathbf{E} = \mathbf{K}'^T \mathbf{F} \mathbf{K}$                                      |

|                    | Fundamental matrix $F$                                                                   | Essential matrix $E$                                                                                       |
|--------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Degrees of freedom | 7 — a $3 \times 3$ matrix, less one for overall scale and one for the rank-2 constraint. | 5 — three for rotation, two for translation direction. Equivalently rank 2 with two equal singular values. |
| Minimum points     | 7 (or 8 for the linear algorithm)                                                        | 5 — <b>and RANSAC cares a great deal</b> , since the sample size sits in the exponent (Module 03 §3).      |
| What you get       | Epipolar lines only.                                                                     | <b>Epipolar lines AND the relative pose</b> , up to scale (§2).                                            |

**Calibration is what converts epipolar geometry into metric pose**, which is the concrete payoff for Module 02's care — and the reason an uncalibrated pipeline can stitch images but cannot measure the scene.

## 2 Pose and triangulation

SVD of  $E$  gives FOUR candidate  $(R, t)$  pairs: two possible rotations, and  $t$  up to sign.

Only ONE of the four puts the triangulated points IN FRONT of both cameras. The other three place points behind one or both, which is physically impossible.

SO: triangulate one correspondence under each candidate and keep the one with positive depth in both views. This is the CHEIRALITY check, and it is two lines of code.

AND  $t$  CARRIES NO SCALE.  $|t| = 1$  by construction, because Module 01's scale ambiguity is a theorem rather than a software limitation. Your reconstruction is correct in SHAPE and arbitrary in SIZE.

TRIANGULATION: two rays that should meet do not, because of noise. Find the 3D point minimising REPROJECTION ERROR in both images -- NOT the midpoint of the rays' closest approach, which is the intuitive answer and is biased.

The noise is in the image plane, not in the world, so the objective belongs in the image plane. The midpoint method weights a distant point's angular error the same as a near one's, which is wrong by a factor of the depth.

**Minimise error where the noise actually is.** That single principle explains the triangulation choice, and **it governs bundle adjustment in Module 05** and the loss function of Module 11's neural reconstruction as well — all three minimise image-plane residuals rather than world-space ones, for the same reason.

### 3 Homographies

#### A HOMOGRAPHY APPLIES IN EXACTLY TWO SITUATIONS

**The scene is planar.** If every point lies on a single plane, the map from one image to the other is a projective transformation of that plane — 8 degrees of freedom, determined by 4 point correspondences. Facades, floors, tabletops, documents, and whiteboards all qualify.

**Or the camera rotates about its centre without translating.** Then depth plays no part in the mapping, because the two rays to any world point differ only by a rotation, and **any scene whatsoever maps by a homography. This is exactly why panorama stitching works** and why it fails if you walk while shooting.

**Both cases are degenerate for the essential matrix** — §4's subject, and the most common failure in practice rather than a theoretical curiosity.

**So estimate both models and compare their inlier counts. If a homography explains the correspondences as well as the essential matrix does, you are in a degenerate or near-degenerate configuration and must not trust the recovered pose. That test costs one extra RANSAC run and almost nobody performs it** — it is the single highest-value check in a two-view pipeline.

### 4 Degeneracies

- **Pure rotation. No baseline means no triangulation is possible and E is undefined**, because  $t$  is zero and  $[t]_{\times}$  vanishes. **The symptom is a plausible rotation with wild, unstable depths** — and it is the most common real failure, because a person photographing a scene from one spot while turning produces exactly this.
- **All points coplanar.** A plane does not determine  $F$  uniquely, so the 8-point algorithm returns an arbitrary member of a family — noise with a confident interface. **Use the homography instead**, which is well-determined here.
- **Very small baseline relative to scene depth.** Not strictly degenerate but badly conditioned, which is Module 01 §2's stability failure in its most practical form. **The depth error scales as depth squared over baseline** (see below).
- **Repeated structure.** Brick walls, office carpet, railings, windows. **Matching is confidently wrong at a consistent offset, so RANSAC finds a large, coherent, incorrect inlier set** — which defeats the usual defence entirely, because the outliers agree with each other.
- **All points at great distance.** Translation becomes unobservable as depth grows, so the configuration tends toward the pure-rotation case continuously rather than abruptly.
- **Every one of these returns a confident answer. Which is why the homography comparison of §3 and a conditioning check on the estimate are mandatory** rather than refinements — nothing else distinguishes a good estimate from a degenerate one.

$$\delta z \approx z^2 \cdot \delta d / (f \cdot b)$$

**DEPTH ERROR GROWS WITH THE SQUARE OF DEPTH**

**For a stereo pair with baseline  $b$ , focal length  $f$  in pixels, and disparity measurement error  $\delta d$ , the resulting depth error is approximately  $Z^2 \delta d / (fb)$ .** The derivation is one line from  $Z = fb/d$ .

**So doubling the distance quadruples the error.** A rig accurate to one centimetre at 2 metres is accurate to about 25 centimetres at 10 metres — which is frequently the difference between a system that works and one that does not, and it is usually discovered after the hardware is built.

**The only levers are baseline and focal length, and both are linear. Widening the baseline improves depth precision and worsens matching**, because the viewpoint change grows and Module 03's unsolved viewpoint invariance starts to bite. **That is the fundamental stereo trade** and Module 06 returns to it.

**Compute this number for your configuration before building anything on it. It decides whether the system is possible at all**, it takes one line, and it is the most useful single calculation in this module.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                      | What it covers for this module                                                                                                  |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Hartley & Zisserman — Multiple View Geometry, chapters 9–12                   | <b>The definitive treatment of this module.</b> Epipolar geometry, F and E, triangulation, and the degeneracies of §4.          |
| Nistér — An Efficient Solution to the Five-Point Relative Pose Problem (free) | <b>The 5-point solver of §1's table</b> , and why the sample size matters so much.                                              |
| Szeliski — Computer Vision, chapter 11 (free PDF)                             | A gentler path through the same material, with the practical pipeline.                                                          |
| Hartley — In Defense of the Eight-Point Algorithm (free)                      | <b>Why normalisation of coordinates matters so much</b> — CSCE 620's and CSCE 669's conditioning lesson, in this exact setting. |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Derive the epipolar constraint from the triple product** and write E in terms of R and t.
- 2 **Implement the 8-point algorithm** with and without coordinate normalisation, and compare the errors.
- 3 **Draw epipolar lines** on a real image pair and verify matches lie on them.
- 4 **Decompose E and implement the cheirality check**, reporting how many points pass under each of the four candidates.

- 5 **Triangulate by midpoint and by reprojection minimisation** and compare the bias on synthetic data with known ground truth.
- 6 **Photograph a scene by pure rotation** and run essential matrix estimation. Report the depths you get.
- 7 **Estimate a homography on the same pair** and compare the inlier counts. Confirm the degeneracy test fires.
- 8 **Photograph a planar facade** and show the 8-point algorithm returns garbage.
- 9 **Compute the depth error formula for your own rig** at 1, 5, and 20 metres.
- 10 **Verify it empirically** by triangulating objects at measured distances.

## Self-check

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Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Derive the epipolar constraint in four lines.
- 2 Why does it reduce search from two dimensions to one?
- 3 Compare F and E on five axes.
- 4 Why does E have five degrees of freedom?
- 5 Describe the four-way decomposition and the cheirality check.
- 6 Why triangulate by reprojection error rather than by midpoint?
- 7 Name the two situations where a homography applies.
- 8 Name five degeneracies and the symptom of each.
- 9 State the depth error formula and its practical consequence.